Hall Ticket Number:

MR-21

Code: B413CS3

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Regular Examinations, Aug/Sep-2023

Database Management Systems

(Common to CSE & IT)

Time: 3 Hours | *Answer ALL Questions* | Max. Marks: 70

Part-A (10 x 2 =20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	What is the Difference between the Candidate Key and Super Key?	2	4	1	1,2,3
1. b)	What is Weak Entity? How it is different from Strong Entity?	2	4	1	1,2
1. c)	Define Referential Integrity. Give an example.	2	2	2	1,2
1. d)	What is View? State its purpose.	2	2	2	1,2
1. e)	What is Functional Dependency? Give an example.	2	1	3	1,2
1. f)	What is Redundancy and why do we want to avoid it when creating a database?	2	4	3	1,2
1. g)	Explain Exclusive Lock. Why we use it?	2	3	4	1,2
1. h)	What are Check Points? What is the use of it?	2	2	4	1,2
1. i)	What do you mean by Indexing?	2	2	5	1,2
1. j)	What is hash File Organization?	2	1	5	1,2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Draw the Architecture of DBMS and explain about its various components.	10	2	1	1,2
	OR				
2. c)	Explain about the functionalities of DBA.	5	2	1	1,2
2. d)	Differentiate between Total Participation and Partial Participation. Illustrate with an example E-R diagram.	5	4	1	1,2
3. a)	Write short notes on below operations in Relational Algebra. Illustrate with an example query in relational algebra. i) Selection ii) Projection iii) Union iv) Cartesian product	10	2	2	1,2
	OR				
3. c)	Consider the following relations: Employee (emp_id, emp_name, gender, salary) Work_For(emp_id, project_id) Project (project_id, project_name, duration_in_months) Write the following queries in SQL. i) Get the details of Employees having salary >10000. ii) Get the names of Employees working for pid = 20. iii) Get the strength of Male and Female Employees. iv) List project_ids in the descending order of their duration.	10	3	2	1,2,3
4. a)	What is Normalization? Why do we need to Normalize our database? Explain 1NF, 2NF, 3NF, BCNF Normal Forms by taking proper examples.	10	2	3	1,2
	OR				
4. c)	Given the Relation R(A,B,C,D) and the set of Functional Dependencies: {**A → BC, B → C, A → B, AB → C, AC → D**}. Find out the irreducible set for the given set of Functional Dependencies.	10	3	3	1,2,3

P.T.O.

5. a)	What is transaction? Give an example. Explain ACID properties in DBMS with the help of examples.	10	2	4	1,2
	OR				
5. c)	Discuss about Log Based Recovery Techniques. Illustrate with an example.	10	2	4	1,2
6. a)	Discuss about organization of records in Sequential File Organization by illustrating with an example after insertion and deletion operations.	5	3	5	1,2
6. b)	Distinguish between Sparse and Dense Index illustrating with an example.	5	4	5	1,2
	OR				
6. c)	Demonstrate insertion, deletion and searching a given element in B+ Trees? Explain with example.	10	3	5	1,2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome

Hall Ticket Number:

MR-21

Code: B413CB3

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Regular Examinations, Aug/Sep-2023

Introduction to Database Management Systems

(Common to CSB, CSM &CSD)

Time: 3 Hours | *Answer ALL Questions* | Max. Marks: 70

Part-A (10 x 2 =20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Write about the Database languages.	2	1	1	1, 2
1. b)	What is a weak entity set? How do you represent it using ER diagram?	2	2	1	1,2
1. c)	What are the SQL statements used to retrieve and modify the database?	2	1	2	1,2
1. d)	What is the difference between primary key and unique key? Give example.	2	3	2	1,2
1. e)	What is schema refinement?	2	1	3	1,2
1. f)	Define multi valued dependency and Join dependency.	2	1	3	1,2
1. g)	What are the properties of Transaction?	2	1	4	1,2
1. h)	When do you say that two schedules are conflict serializable and View serializable?	2	2	4	1,2
1. i)	Differentiate between Primary Indexing and Secondary Indexing.	2	3	5	1,2
1. j)	Write the differences between tree based and Hash based indexes.	2	3	5	1,2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	What is DBMS? Explain its advantages over file system.	5	1	1	1,2
2. b)	What are speciality databases? Explain.	5	1	1	1,2
	OR				
2. c)	Draw an ER diagram for an Airline Reservation system.	5	3	1	1,2
2. d)	Write about the Integrity constraints over Relations with examples.	5	1	1	1,2
3. a)	Explain the fundamental operations in relational algebra with examples.	5	1	2	1,2
3. b)	What is a view? Can you create view on multiple tables? Illustrate With examples.	5	3	2	1,2
	OR				
3. c)	Consider the following schema: Suppliers(sid: integer, sname: string, address: string) Parts(pid: integer, pname: string, color: string) Catalog(sid: integer, pid: integer, cost: real) The Catalog relation lists the prices charged for parts by Suppliers. Write the following queries in SQL 1. Find the pnames of parts for which there is some supplier. 2. Find the snames of suppliers who supply every part. 3. Find the pnames of parts supplied by Acme Widget Suppliers and by no one else. 4. Find the sids of suppliers who charge more for some part than the average cost of that part (averaged over all the suppliers who supply that part). 5. For each part, find the sname of the supplier who charges the most for that par.	5	3	2	1,2
3. d)	What is trigger? Explain how to implement triggers in SQL?	5	3	2	1,2

P.T.O.

4. a)	Explain in detail about 1NF, 2NF and 3NF with suitable examples.	5	2	3	1,2
4. b)	Find the highest normal form in R(A, B, C, D, E) under following functional dependencies. ABC -> D CD -> AE.	5	3	3	1,2,3
	OR				
4. c)	What are the properties of Decomposition. Explain.	5	3	3	1,2
4. d)	Suppose that we decompose the schema R = (A, B, C, D, E) into R1 (A, B, C) and R2 (A, D, E). Determine that this decomposition is a lossless-join decomposition or dependency preserving if the following set F of functional dependencies holds: A → BC, CD → E,B → D, E → A.	5	3	3	1,2,3
5. a)	Explain Lock based protocols.	5	1	4	1,2
5. b)	How do you implement atomicity property of transactions?	5	2	4	1,2
	OR				
5. c)	Explain the concept of Serializability.	5	1	4	1,2
5. d)	How do you test for the Serializbility property of transactions?	5	2	4	1,2
6. a)	How does Extendable hashing use a directory of buckets? How does it handle the insert and delete operations?	5	1	5	1,2
6. b)	Compare and contrast Hash based indexing and tree based indexing.	5	2	5	1,2
	OR				
6. c)	Describe the Insertion and Deletion Operations in B+ trees with an example.	5	3	5	1,2
6. d)	Make a comparison between the primary index and a secondary index.	5	2	5	1,2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome